Ericsson PE Node Troubleshooting SOP for Optical Transport Failure ALARM_9195 at Indiranagar Exchange

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_9195

Description

The Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the PE Node (IND-PE-03) at the Indiranagar Exchange is attributed to the degradation of the optical signal quality due to the physical degradation of the SFP/XFP transceivers or the fiber infrastructure. The Ericsson PE node operates in an optical transport network, transmitting and receiving data through optical signals. The degradation of the SFP/XFP transceivers or the fiber cuts can result in a decrease in the received power, leading to a signal quality degradation. This can be exacerbated by the accumulation of physical stress, temperature fluctuations, and optical signal attenuation in the fiber infrastructure. The ALARM_9195, indicating a minor optical transport failure, signifies the system's attempt to maintain communication despite the signal quality degradation. To troubleshoot this issue, we will perform diagnostic checks to identify the root cause of the signal quality degradation and undertake resolution steps to rectify the issue.

Pre-requisite Checks

- Prerequisite: Verify the site accessibility and availability of necessary physical access to the PE Node at Indiranagar Exchange., Rationale: Ensure that the remote hands are dispatched to the site to facilitate the troubleshooting process.

- Prerequisite: Check the redundant link status to confirm that the secondary link is functioning properly., Rationale: Verify the redundancy of the optical transport network to ensure continuous communication in case of a failure.

- Prerequisite: Gather relevant equipment logs to analyze the system's behavior and identify any potential patterns or anomalies., Rationale: Analyze the historical data to determine the potential root causes of the failure.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP transceiver	High
Fiber cuts or physical damage	Medium
Rx/Tx power loss due to signal attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show optic sfp , Expected Output: SFP Slot - Rx/Tx Power:
dBm, Signal Quality: , Description: This command checks the SFP/XFP
transceiver status, including the received and transmitted power
levels and signal quality.

Command: show optic fiber , Expected Output: Fiber ID - Attenuation:
dB, Loss: dB, Description: This command checks the fiber
infrastructure status, including the attenuation and loss values.

Command: show optic signal , Expected Output: Signal ID - Power: dBm,
Quality: , Description: This command checks the signal quality and
power levels, helping to identify potential issues with the optical
transport network.
```

Resolution Steps

Step 1: Step: Replace the degraded SFP/XFP transceiver, Command: replace optic sfp with , Description: Replace the degraded SFP/XFP transceiver with a new one to restore the optical signal quality.

Step 2: Step: Repair or replace the damaged fiber infrastructure, Command: replace optic fiber with , Description: Repair or replace the damaged fiber infrastructure to restore the optical signal quality.

Step 3: Step: Adjust the Rx/Tx power levels to compensate for signal attenuation, Command: set optic signal power dBm, Description: Adjust the received and transmitted power levels to compensate for signal attenuation and restore the optical signal quality.

Post-Check Verification

```
Command: show optic sfp , Expected Output: SFP Slot - Rx/Tx Power:
dBm, Signal Quality: , Description: Verify that the SFP/XFP
transceiver status has been restored to normal.

Command: show optic fiber , Expected Output: Fiber ID - Attenuation:
dB, Loss: dB, Description: Verify that the fiber infrastructure status
has been restored to normal.

Command: show optic signal , Expected Output: Signal ID - Power: dBm,
Quality: , Description: Verify that the signal quality and power
levels have been restored to normal.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: replace optic sfp with , Description: Restore the original SFP/XFP transceiver to its previous state if the replacement fails.

- Step: Restore the original fiber infrastructure, Command: replace optic fiber with , Description: Restore the original fiber infrastructure to its previous state if the replacement fails.

- Step: Restore the original Rx/Tx power levels, Command: set optic signal power dBm, Description: Restore the original received and transmitted power levels if the adjustment fails.